

SMART INDIA HACKATHON 2026

Deep Learning-Based Super-Resolution Mapping (SRM) from
Medium-Resolution Satellite Imageries

GeoSR-AI

Uncertainty-aware super-resolution of Sentinel-2 imagery: 10 m to a 2.5 m grid

HIGHER RESOLUTION | UNCERTAINTY-AWARE | REAL-WORLD READY

Table 1. Submission details.

Field	Value
Problem Statement ID	SIH 26142
Problem Statement	Deep Learning-Based Super-Resolution Mapping (SRM) from Medium-Resolution Satellite Imageries
Theme / PS Category	Space Technology / Software
Team ID / Team Name	178177 / PlotSphere
Solution	GeoSR-AI: two-stage EDSR with validated uncertainty, out-of-range guard and GeoTIFF output

+1.48 dB

PSNR over bicubic, held-out
Indian data (100% of 238
patches)

+0.030

real sub-10 m detail on unseen
2.5 m references (CI
0.013-0.046)

$\rho = 0.50$

uncertainty vs real error; top
10% of pixels carry ~4× the error

95.5 s

10 km × 10 km scene to a 2.5 m
GeoTIFF on a laptop RTX 3050

Abstract

Medium-resolution Sentinel-2 imagery (10 m) is freely available every few days, but it is too coarse to resolve narrow roads, field boundaries and small structures. We present **GeoSR-AI**, a super-resolution framework that converts the four 10 m Sentinel-2 bands (B02, B03, B04, B08) to a 2.5 m grid (×4), preserves the geolocation and spectral balance of the input, and delivers a per-pixel uncertainty map with every product. A residual CNN (EDSR, 1.52 M parameters) is first trained on 2,265 Indian Sentinel-2 patches (4 cities × 5 dates) with physics-motivated, cloud-masked losses, then jointly fine-tuned on real Sentinel-2/2.5 m reference pairs from the opensr-test benchmark while continuing to see Indian data.

On spatially held-out Indian data the final model improves on bicubic interpolation by **+1.48 dB PSNR [95% CI 1.44, 1.52]** on 100% of patches, with 19% lower spectral-angle error and 22% lower NDVI error. On real 2.5 m references from sensors and locations never seen in training it recovers **measurable real sub-10 m detail (+0.030 detail correlation, CI 0.013–0.046, 79% win rate; 100% on WorldStrat SPOT)** while radiometric PSNR stays on par with bicubic. Its test-time-augmentation uncertainty correlates with the true error (Spearman $\rho = 0.43$ synthetic, **0.50 on real references**). We also report what did not work: adversarial (GAN) training traded accuracy for invented texture, and a saturation failure on bright roofs was found and fixed with an out-of-range guard.

1. PROBLEM AND OBJECTIVES

Sentinel-2 offers global, 5-day, free multispectral imagery, which makes it the default data source for crop monitoring, land-cover mapping, urban growth analysis and disaster response in India. At 10 m, however, a two-lane road, a field bund or a small building occupies at most one pixel, which limits interpretation. The problem statement asks for a robust deep-learning framework that enhances such imagery to a finer (<4 m) product, preserves geospatial and spectral consistency, validates the result against high-resolution references, and clearly accounts for uncertainty because some reconstructed detail is inferred rather than observed.

Terminology. In remote sensing, “super-resolution mapping” classically denotes sub-pixel land-cover mapping [12]. Following the problem statement, we use SRM for the enhancement of the multispectral image itself (single-image super-resolution); downstream maps such as NDVI are derived from the enhanced bands.

Objectives. (1) A $\times 4$ super-resolution model for Sentinel-2 B02/B03/B04/B08 that outputs a georeferenced 2.5 m product; (2) spectral fidelity measured explicitly (SAM, ERGAS, NDVI error); (3) leak-free validation on unseen Indian cities and on **real** 2.5 m references; (4) a per-pixel uncertainty layer whose reliability is itself measured; (5) an end-to-end, tile-based inference tool for full scenes on a consumer GPU.

Contributions. A two-stage training strategy (synthetic Indian data $\rightarrow$ joint synthetic + real fine-tuning) that is the only configuration in our study to improve both benchmarks; a calibrated uncertainty map validated against real error; an out-of-range guard that removes a saturation artefact; and an honest ablation showing why a GAN was rejected.

2. DATA

2.1 Sentinel-2 training data (India)

We use the four 10 m Sentinel-2 bands B02 (blue), B03 (green), B04 (red) and B08 (near-infrared) over four Indian cities with contrasting climates, five acquisition dates each (January 2025 – June 2026). Scenes were tiled into non-overlapping 512×512 -pixel patches ($5.12 \text{ km} \times 5.12 \text{ km}$, stride 512 px). Opaque clouds and no-data are stored as NaN and excluded from every loss and metric through a validity mask; patches below 90% valid pixels are not used (observed validity: min 90.1%, mean 99.7%, median 100%).

Table 2. Dataset composition (source: masterpatchindex.csv).

City	Dates (scene 1–5)	Patches per date	Total	Unique locations
Ahmedabad	2025-05-01, 04-03, 05-16, 03-22, 04-06	108 / 108 / 107 / 108 / 108	539	216
Guwahati	2025-01-10, 02-09, 2026-02-19, 03-08, 03-11	45 / 48 / 24 / 48 / 27	192	72
Nashik	2025-01-28, 02-27, 04-28, 03-29, 2026-06-17	65 / 46 / 80 / 80 / 36	307	80
New Delhi	2025-06-12, 06-07, 05-23, 05-13, 04-23	255 / 218 / 255 / 252 / 247	1,227	255
Total	20 scenes		2,265	623

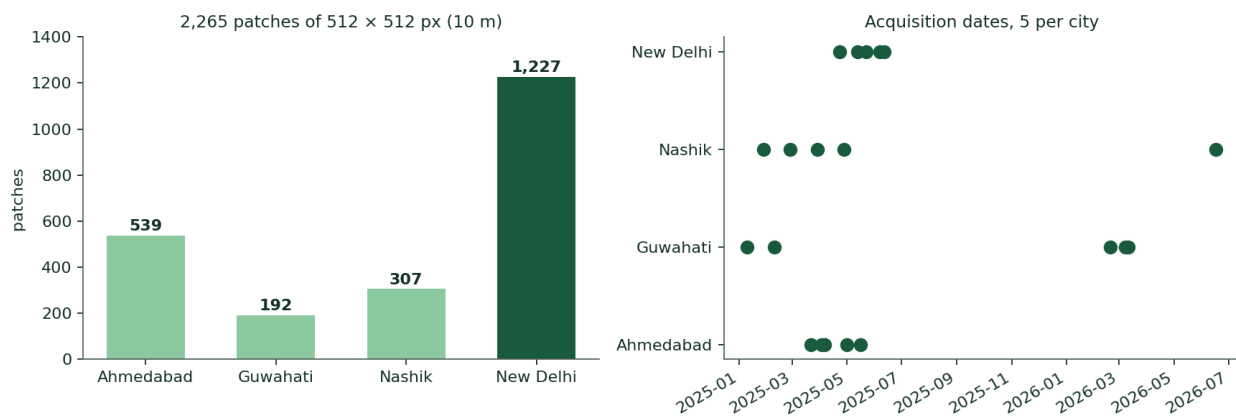


Figure 1. Patch counts per city (stacked by acquisition date) and acquisition timeline. New Delhi contributes 54% of the data; the test-city choice in Section 3.2 accounts for this imbalance.

Acquisition and pre-processing. Level-2A (surface reflectance) Sentinel-2 products were obtained from the Copernicus Data Space Ecosystem for the four study cities. A pre-processing notebook tiles each scene into 512×512 patches stored as (4, H, W) float32 arrays in band order B02, B03, B04, B08, and writes a metadata file per scene (patch id, row and column offset, patch size, valid-pixel percentage). Opaque clouds and no-data are stored as NaN, are excluded from every loss and metric, and patches with less than 90% clear pixels are dropped. The digital numbers follow processing baseline ≥ 04.00 (reflectance $\times 10,000 + 1,000$); this was verified independently: the band means of a separately downloaded L2A scene (Nashik, 4 June 2026) matched the training statistics within $0.98\text{--}1.03\times$.

Consolidation and quality control. A consolidation script merges the 20 per-scene metadata files into one master index, removes entries whose array is missing on disk and assigns a global id. A validation script then samples patches from every city and flags: a band count other than 4, more than 15% NaN, maximum values above 20,000 (wrong scaling), suspiciously low values (double scaling) and negative reflectance in any band. A further integrity check in the training code verifies that every indexed file exists, reports shapes and NaN fractions, and confirms zero train/validation footprint overlap.

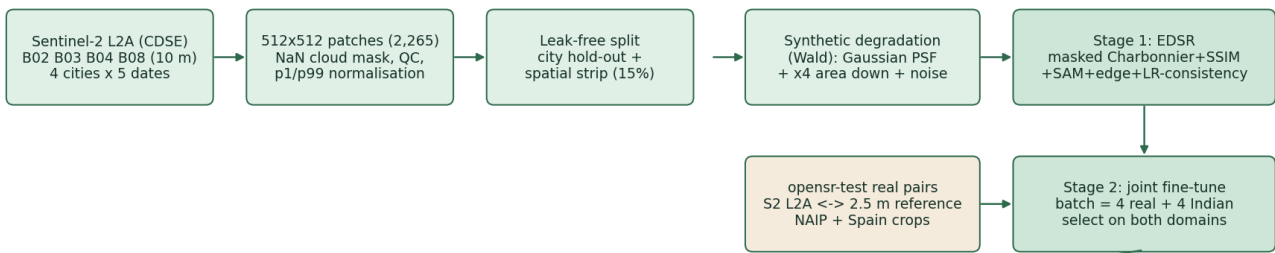
Normalisation. Each band is scaled with shared percentile statistics (p1/p99: B02 1266/3205, B03 1459/3512, B04 1326/3945, B08 2050/5268 DN; computed over 473.6 M valid pixels). The final model clips normalised values to [0, 1.5]; values above this range are handled by the guard described in Section 3.6.

2.2 Real high-resolution reference data

For validation at the target scale we use **opensr-test** [3], a benchmark that pairs real Sentinel-2 L2A images with co-registered, radiometrically harmonised 2.5 m references ($\times 4$): NAIP (USA, 62 aerial images), SPOT 6/7 from WorldStrat [4] (9 satellite images), Spain crops (28) and Spain urban (20). NAIP and Spain crops are used for fine-tuning; **SPOT and Spain urban (29 images) are never used in training** and form the real test set. Its L2A values carry no +1,000 offset, so the offset is added before our normaliser.

3. METHODOLOGY

A. Data & training



B. Inference (any Sentinel-2 scene)

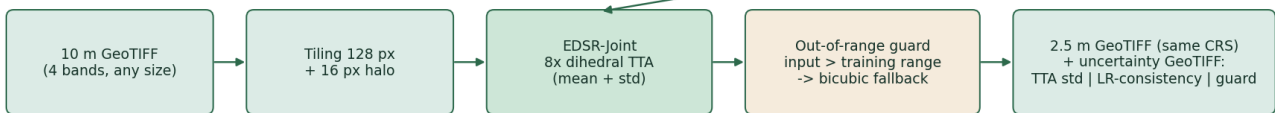


Figure 2. End-to-end framework: data preparation and two-stage training (A) and scene-level inference with uncertainty outputs (B).

3.1 Synthetic training pairs (Wald protocol)

Because no Indian 2.5 m imagery is available, stage-1 pairs follow Wald’s protocol [2]: each 10 m patch is the target and its input is synthesised by a Gaussian point-spread function followed by $\times 4$ area downsampling (10 m $\rightarrow$ 40 m-equivalent). During training the PSF width is sampled from $\sigma \in [0.6, 1.2]$ HR pixels and additive noise from 0–1% of the band standard deviation, for robustness to the unknown real sensor MTF; validation and test use a fixed $\sigma = 0.9$ without noise. A single implementation of this degradation is shared by the data loader, the consistency loss, the uncertainty map and inference, so all components assume the same sensor model.

3.2 Leak-free data splits

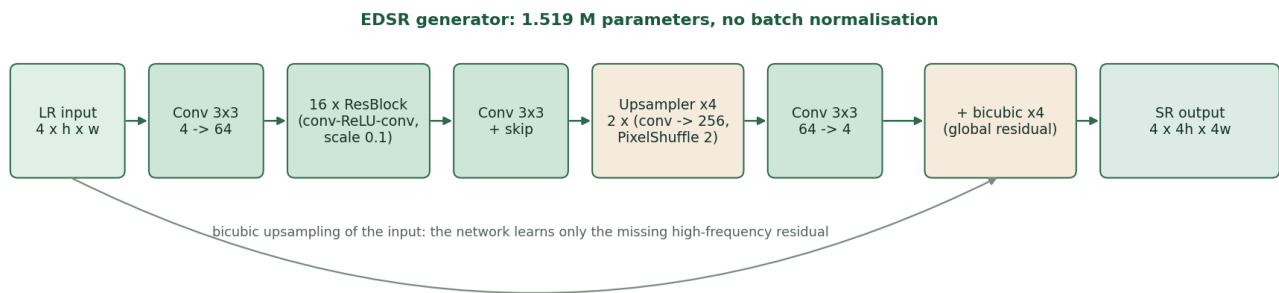
The five dates of a city image the same ground, so a random or per-date split would leak geography. We therefore use (i) **city hold-out** for generalisation tests and (ii) a **spatial strip** for validation: the last 15% of each training city’s extent (all dates) becomes validation, and training patches whose footprint overlaps the strip are discarded. A footprint check confirmed zero train/validation overlap. Nashik (307 patches) was held out instead of New Delhi (54% of all data) so that training data was not halved.

Table 3. Splits used in this work.

Regime (use)	Train	Validation	Test	Dropped (overlap)
Test city = Nashik (generalisation, GAN study)	1,638 (Ahm 450, Guw 168, Del 1,020)	202	307 (Nashik)	118
Test city = Guwahati (generalisation)	1,707 (Ahm 450, Nsk 237, Del 1,020)	220	192 (Guwahati)	146
All cities (final model)	1,875 (all four cities)	238 (Ahm 45, Guw 18, Nsk 36, Del 139)	—	152

3.3 Model

The generator is an EDSR network [1] without batch normalisation: a 3×3 head convolution (4 → 64 channels), 16 residual blocks (residual scaling 0.1), two PixelShuffle ×2 stages [13], and a 3×3 tail convolution back to 4 bands. Its output is **added to a bicubic upsampling of the input** (global residual), so the network only learns the missing high-frequency detail and the low-frequency radiometry stays anchored to the observed image. The model has 1.519 M parameters (6.1 MB checkpoint). The output is not clamped during training, which avoids zero gradients at saturation.

**Figure 3.** EDSR generator with a global bicubic residual connection.

3.4 Loss function

All terms are masked by the validity mask so cloud gaps never act as ground truth: $L = 1.0 \cdot \text{Charbonnier} + 0.1 \cdot (1 - \text{SSIM}) + 0.05 \cdot \text{SAM} + 0.05 \cdot \text{Sobel-gradient} + 0.1 \cdot \text{LR-consistency}$. The Charbonnier term [16] drives pixel fidelity, SSIM [6] structure, the spectral-angle term [10] preserves band ratios (and therefore NDVI), the gradient term sharpens edges such as roads and field boundaries, and the LR-consistency term requires the output, re-degraded with the sensor model, to reproduce the observed input.

3.5 Two-stage training

Table 4. Training configuration (from saved configs and checkpoints).

Setting	Stage 1 – synthetic (EDSR-Synthetic)	Stage 2 – joint fine-tune (EDSR-Joint, final)
Initialisation	random	Stage-1 all-city model
Data	Indian patches, synthetic ×4 pairs	Each batch: 4 real opensr-test pairs (NAIP + Spain crops) + 4 synthetic Indian patches
Crops	128×128 HR (32×32 LR), 4 random crops per patch per epoch	128×128 HR, 16 random crops per real image per epoch
Augmentation	8 rotations/flips	8 rotations/flips
Optimiser / LR	Adam (0.9, 0.99), 2×10^{-4} , 500-step warm-up + cosine	Adam (0.9, 0.99), 5×10^{-5} , warm-up + cosine
Batch / epochs	8 / 60 (early-stopping patience 12)	8 / 30 (patience 8)
Precision / clipping	mixed precision, grad-norm 1.0	mixed precision, grad-norm 1.0
Normalisation clip	[0, 1] (city-hold-out runs); [0, 1.5] (final)	[0, 1.5]
Model selection	best validation PSNR (spatial strip)	sum of PSNR gains on 14 held-out real images and 64 Indian validation patches

Stage 2 addresses a failure observed in development: fine-tuning on real pairs alone raised real-data detail but erased two thirds of the Indian gain (Section 5.3). Mixing both domains in every batch, and selecting the checkpoint on both, keeps the Indian prior while learning the real Sentinel-2 $\rightarrow$ 2.5 m relationship.

3.6 Inference, uncertainty and the out-of-range guard

- **Tiled inference.** Scenes are processed in 128-pixel tiles with a 16-pixel halo and written window-by-window to a GeoTIFF on a 2.5 m grid with the input CRS and a rescaled transform, so any scene size fits in 4 GB of GPU memory.
- **Uncertainty (TTA spread).** The input is passed in all 8 rotations/flips; the de-transformed outputs are averaged (final prediction) and their standard deviation is the uncertainty. Disagreement marks detail that is not constrained by the input.
- **LR-consistency error.** $|\text{degrade}(\text{SR}) - \text{input}|$ shows where the output contradicts what the satellite measured.
- **Out-of-range guard.** Where any input band exceeds the normalisation range the model was trained on (> 1.5), the output falls back to bicubic of the unclipped input and the pixel is flagged. Without it, bright roofs saturate band-by-band and turn pink (Figure 8). The guard affected about 2% of pixels on the real test set.

Each product therefore consists of a 4-band 2.5 m image and a 3-band uncertainty GeoTIFF (TTA std, LR-consistency error, guard mask).

3.7 Adversarial variant (ablation)

To test a generative approach, the stage-1 model was fine-tuned with a spectral-normalised PatchGAN discriminator [14, 15] and a relativistic average loss [7, 8] (adversarial weight 5×10^{-3} or 1×10^{-3} ; generator LR 5×10^{-5} , discriminator LR 1×10^{-4} ; 15 epochs).

4. EVALUATION PROTOCOL

- **Synthetic benchmark (40 m $\rightarrow$ 10 m).** Full 512 $\times$ 512 test patches, fixed degradation, 8 $\times$ TTA. PSNR and SSIM on normalised values; SAM (degrees) [10], ERGAS [11] and NDVI mean absolute error on reflectance after removing the +1,000 DN offset. All metrics are masked, computed per image and then averaged.
- **Real benchmark (10 m $\rightarrow$ 2.5 m).** PSNR/SSIM after per-image percentile normalisation of the reference; SAM on reflectance; **detail correlation**, the correlation between the high-frequency layers (image minus Gaussian blur, $\sigma = 2$ px) of the output and the real reference, which directly tests whether added detail is real; LR-consistency; geolocation shift by phase correlation. A 16-pixel border is excluded.
- **Statistics.** 95% bootstrap confidence intervals (2,000 resamples); paired gains over bicubic on identical images; win rate = share of images where the model beats bicubic.
- **Uncertainty reliability.** Spearman ρ between TTA standard deviation and absolute error over valid pixels, and the error ratio between the most- and least-uncertain deciles.

Absolute PSNR values are not comparable between the synthetic and real tables (different normalisations and targets); only gains within a table are compared.

5. RESULTS

5.1 Generalisation to unseen Indian cities

Table 5. Held-out-city results, stage-1 EDSR vs bicubic (synthetic 40 m $\rightarrow$ 10 m, 8 $\times$ TTA). Arrows: $\uparrow$ higher is better, $\downarrow$ lower is better.

Test city (patches)	Method	PSNR $\uparrow$	SSIM $\uparrow$	SAM $^\circ \downarrow$	ERGAS $\downarrow$	NDVI MAE $\downarrow$	Gain vs bicubic (95% CI), win rate
Nashik (307)	Bicubic	23.65	0.604	2.85	3.42	0.0485	—
	EDSR	24.90	0.702	2.37	2.96	0.0393	+1.249 dB (1.218, 1.279), 100%
Guwahati (192)	Bicubic	31.66	0.773	1.56	1.72	0.0245	—
	EDSR	32.38	0.811	1.44	1.57	0.0223	+0.715 dB (0.681, 0.752), 100%

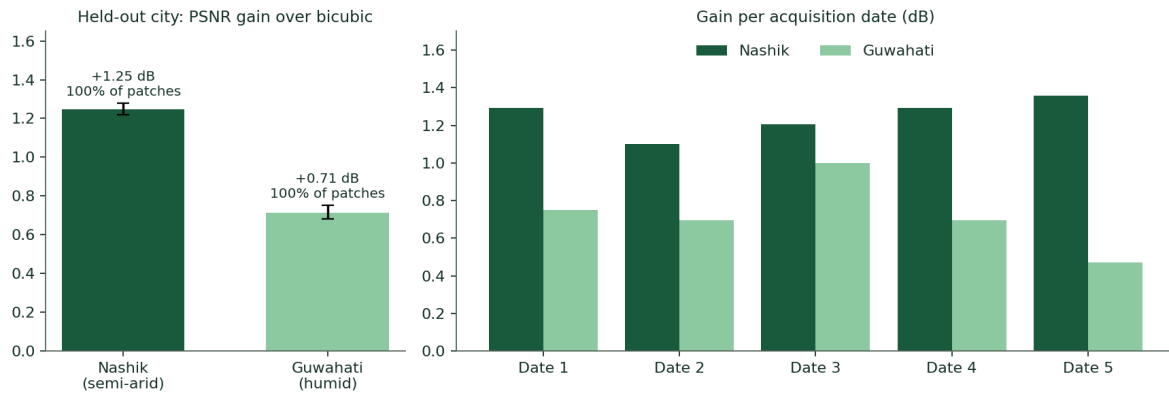


Figure 4. Gain over bicubic on the two held-out cities, overall and per acquisition date. Every patch of both cities improves.

The model improves every patch of two cities it never saw. The gain is smaller in humid, hilly Guwahati than in semi-arid Nashik. Guwahati scene 5 (2026-03-11) is visibly haze-affected (bicubic already scores 37.2 dB because little detail survives), but excluding it only moves the Guwahati gain from +0.72 to +0.75 dB, so the gap is attributed mainly to climate and terrain domain shift rather than data quality. The near-infrared band gains +1.28 dB (Nashik) and +0.70 dB (Guwahati).

5.2 Why a GAN was rejected

Table 6. Adversarial fine-tuning on the unseen Nashik city (adversarial weight 1×10^{-3}).

Model	PSNR $\uparrow$	SSIM $\uparrow$	SAM $^\circ \downarrow$	NDVI MAE $\downarrow$	Gain vs bicubic (95% CI)	Win rate
Bicubic	23.65	0.604	2.85	0.0485	—	—
EDSR (stage 1)	24.90	0.702	2.37	0.0393	+1.249 (1.218, 1.279)	100%
GAN, epoch 1	24.65	0.688	2.43	0.0404	+1.007 (0.977, 1.036)	100%
GAN, epoch 15	24.19	0.659	2.62	0.0436	+0.540 (0.505, 0.575)	94.8%

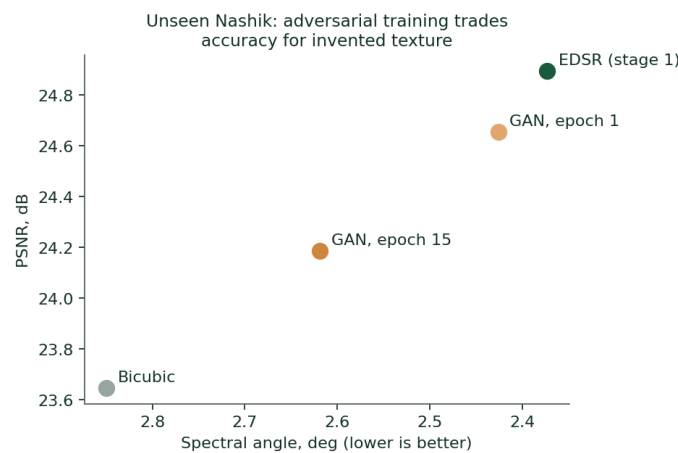


Figure 5. Perception–distortion trade-off [9]: the longer the adversarial training, the lower both radiometric and spectral accuracy.

Every metric degrades monotonically with adversarial training. Visually the GAN output is crisper, but it draws parcel outlines that do not match the reference, and its TTA uncertainty rises exactly there. With a stronger adversarial weight (5×10^{-3}) validation PSNR fell below bicubic (-0.35 dB) by epoch 15 and training was stopped. Because the outputs are meant for quantitative use (NDVI, classification), the non-adversarial model was retained; the uncertainty map still flagged the invented detail (TTA $\rho = 0.44$ and LR-consistency $\rho = 0.29$ for the GAN).

5.3 Final model selection: synthetic vs real vs joint

Zero-shot tests of the synthetic-only model on all 119 real opensr-test images showed a PSNR tie with bicubic (-0.04 dB, CI -0.11 to $+0.03$) and only a small real-detail gain ($+0.011$, CI 0.005 – 0.016), concentrated on NAIP farmland ($+0.024$, 89% of images). This motivated stage 2. Table 7 compares the three candidates on both benchmarks with identical settings (guard on, $8 \times$ TTA).

Table 7. Final comparison. Left: Indian spatial hold-out (238 patches, synthetic). Right: real 2.5 m references never used in training (SPOT + Spain urban, 29 images).

Model	India PSNR gain (dB)	India SAM ° / NDVI MAE	Real PSNR gain (dB)	Real detail-corr. gain	Real SAM °	Uncert. p (real)
Bicubic	0	2.70 / 0.0457	0	0	3.19	—
EDSR-Synthetic (stage 1)	+1.540 (1.501, 1.584)	2.15 / 0.0353	−0.217 (−0.426, −0.060)	−0.000 (−0.010, 0.009)	3.60	0.496
EDSR-Real-FT (real only)	+0.595 (0.562, 0.629)	2.52 / 0.0419	+0.190 (−0.036, 0.381)	+0.038 (0.020, 0.056)	3.17	0.497
EDSR-Joint (final)	+1.479 (1.441, 1.524)	2.18 / 0.0357	+0.004 (−0.224, 0.196)	+0.030 (0.013, 0.046)	3.38	0.500

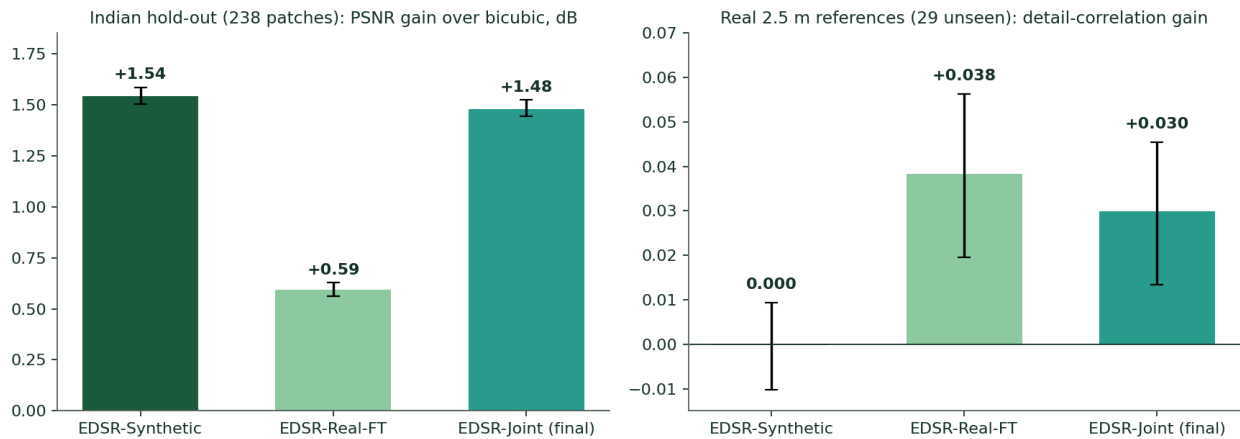


Figure 6. Each single-domain model wins only in its own domain; the joint model is the only one significantly better than bicubic on both benchmarks.

The joint model keeps **96% of the synthetic gain** (1.48 vs 1.54 dB) and **79% of the real-detail gain** (0.030 vs 0.038), and it is the only candidate whose confidence intervals exclude zero on both key metrics. The cost relative to real-only fine-tuning is about 0.19 dB of real PSNR and 0.2° of SAM. On real data the benefit appears as structure (SSIM 0.388 vs 0.348 for bicubic; detail correlation 0.298 vs 0.268) rather than lower pixel error, which is expected when fine detail was never observed by the sensor. No measurable geolocation shift was introduced (0.002 of a 10 m pixel).

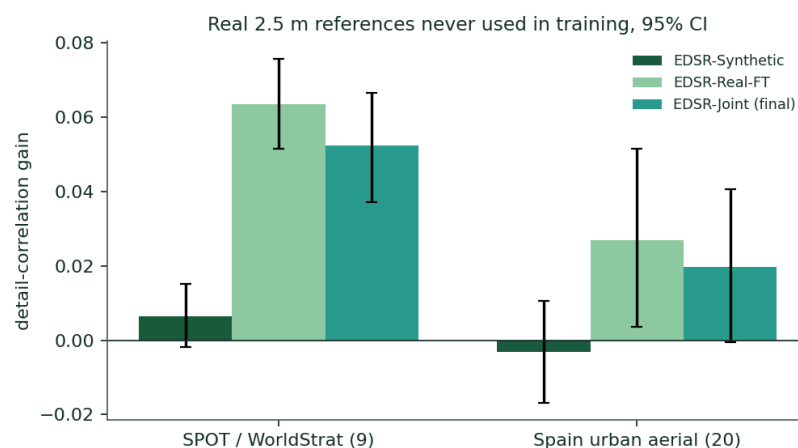


Figure 7. Real-reference detail gain per dataset. On WorldStrat SPOT the final model beats bicubic on 100% of images (+0.052, CI 0.037–0.067); on dense Spanish urban scenes the gain is +0.020 (CI −0.001 to 0.041), borderline.

5.4 Failure analysis: saturation on bright roofs

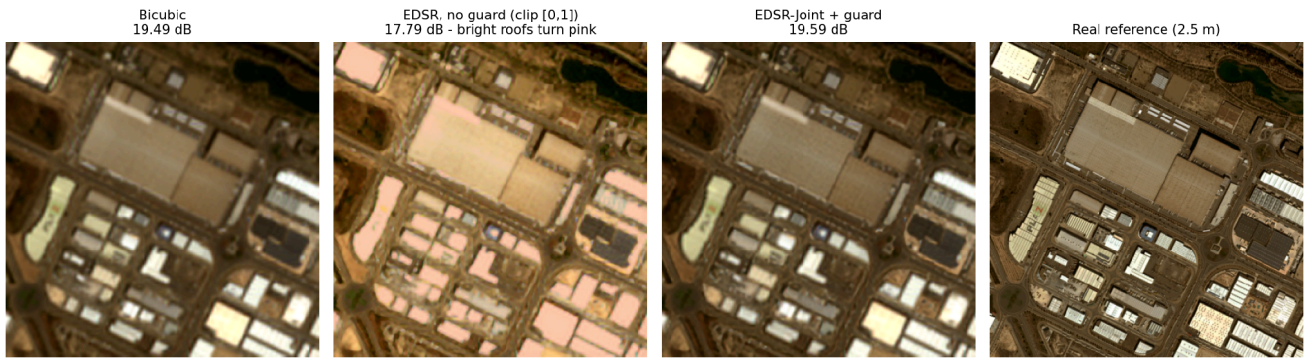


Figure 8. Real test image (Spain urban #0). Without the guard, bright roofs exceed the training range and clip band-by-band (blue first), turning pink and losing 1.7 dB; with the guard the final model beats bicubic (19.59 vs 19.49 dB).

The artefact appeared in all models trained with a fixed clip range. Widening the clip to $[0, 1.5]$ and adding the guard removed it while replacing only $\approx 2\%$ of output pixels (1.99% SPOT, 2.10% Spain urban) with bicubic; the guard mask is exported so users know where the model abstained.

5.5 Uncertainty that can be trusted

Table 8. Reliability of the TTA uncertainty map.

Setting	Model	Spearman ρ (TTA std vs error)	Error ratio top/bottom decile
Nashik, unseen city (synthetic)	EDSR stage 1	0.415	3.9×
Guwahati, unseen city (synthetic)	EDSR stage 1	0.353	3.1×
India spatial hold-out (synthetic)	EDSR-Joint	0.432	3.9×
Real 2.5 m: SPOT / Spain urban / all	EDSR-Joint	0.585 / 0.449 / 0.500	—

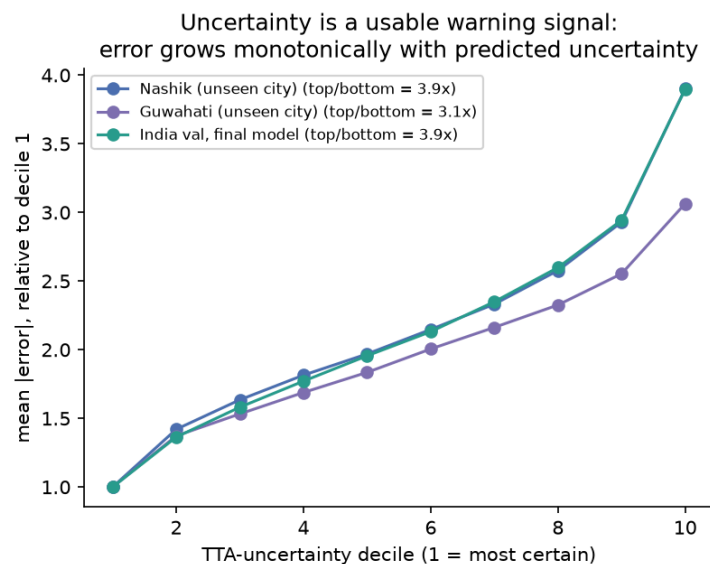


Figure 9. Mean absolute error rises monotonically with predicted uncertainty in every setting.

The uncertainty ranking is consistent across cities, models and, importantly, real references ($\rho \approx 0.50$ for all three candidates), so it is a property of the method rather than of one checkpoint. The LR-consistency map is a weaker error predictor ($\rho = 0.13\text{--}0.20$ synthetic) and is reported as a physical-consistency check rather than as uncertainty.

5.6 Spectral behaviour and training dynamics

Table 9. Per-band PSNR on the Indian spatial hold-out (EDSR-Joint).

Band	B02 blue	B03 green	B04 red	B08 NIR
Bicubic (dB)	24.54	24.00	23.19	23.10

Band	B02 blue	B03 green	B04 red	B08 NIR
EDSR-Joint (dB)	25.94	25.41	24.81	24.59
Gain (dB)	+1.39	+1.41	+1.62	+1.49

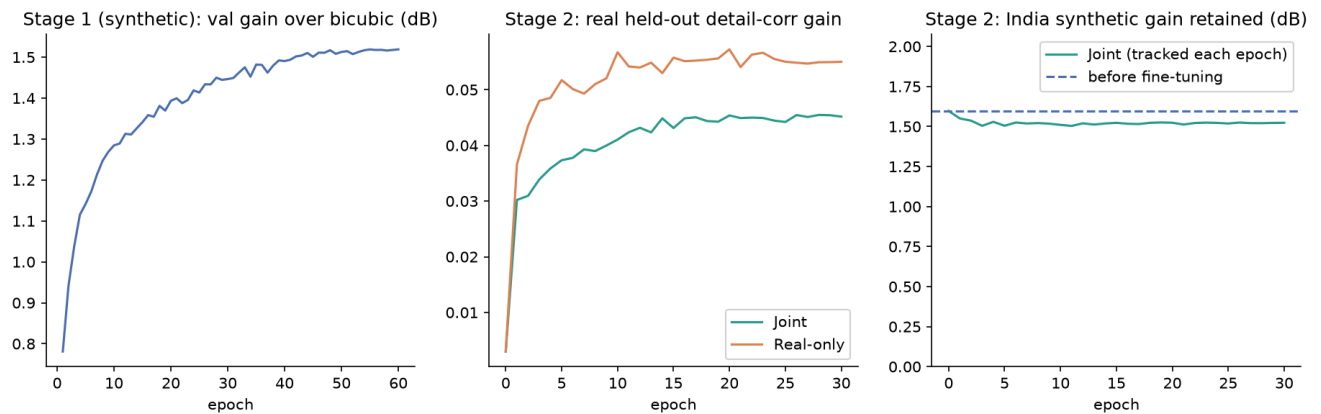


Figure 10. Left: stage-1 validation gain converges to about +1.5 dB in 60 epochs. Middle/right: during joint fine-tuning the real held-out detail gain rises while the Indian gain stays near +1.52 dB (dashed: before fine-tuning).

5.7 Qualitative results

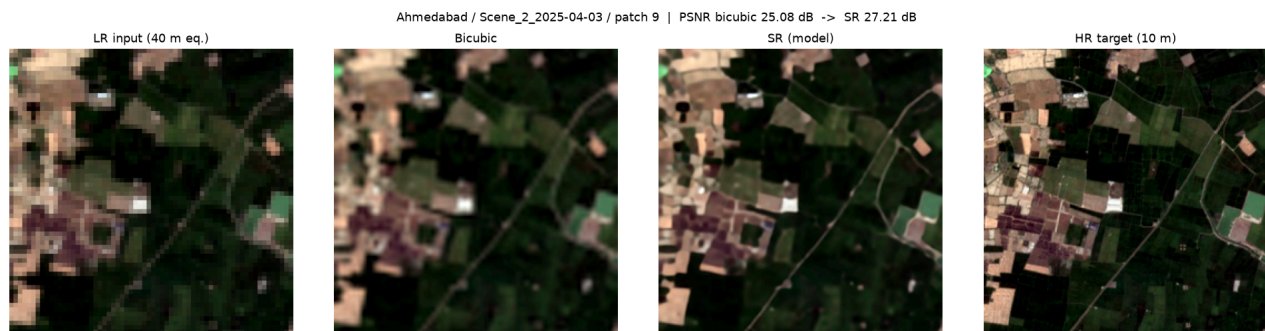


Figure 11. Indian spatial hold-out (Ahmedabad): 40 m-equivalent input, bicubic, EDSR-Joint and 10 m target. Roads, parcel boundaries and the settlement edge are recovered.

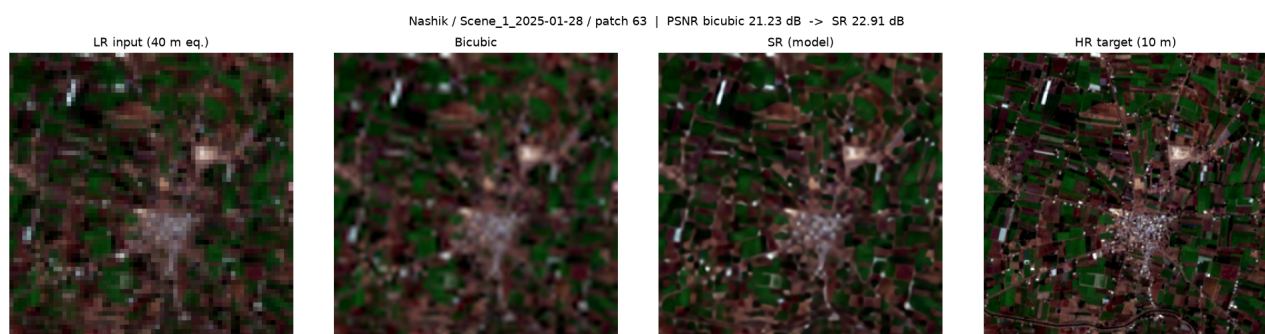


Figure 12. Indian spatial hold-out (Nashik): field network and village; sub-pixel buildings remain smoothed.

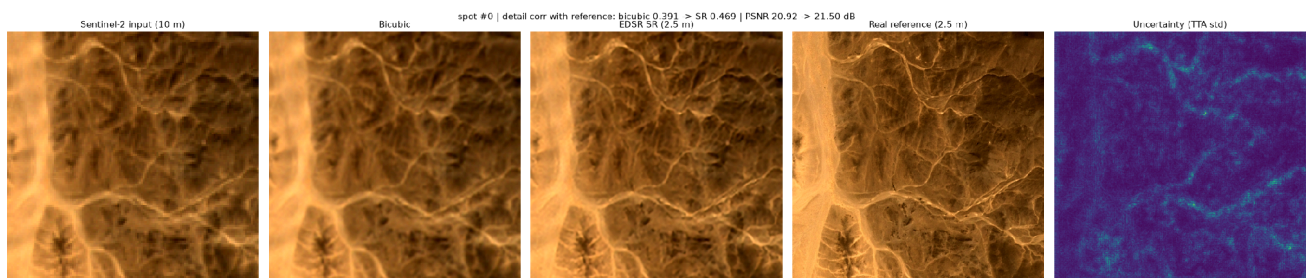


Figure 13. Real WorldStrat SPOT reference (never used in training): the final model raises detail correlation from 0.391 to 0.469 and PSNR from 20.92 to 21.50 dB; uncertainty concentrates on drainage lines.

5.8 Full-scene demonstration: Nashik, 4 June 2026

A real Sentinel-2 L2A scene (tile T43QCC, 2026-06-04) was cropped to 10 km × 10 km (1,000 × 1,000 px) and processed end-to-end to a 4,000 × 4,000 px, 2.5 m GeoTIFF in EPSG:32643 in **95.5 s** with 8× TTA on a laptop RTX 3050 (including model loading). The scene's band means were within 0.98–1.03× of the training statistics (farmland crop: 0.87–0.93×). There is no 2.5 m reference for this scene, so it is a qualitative demonstration only.

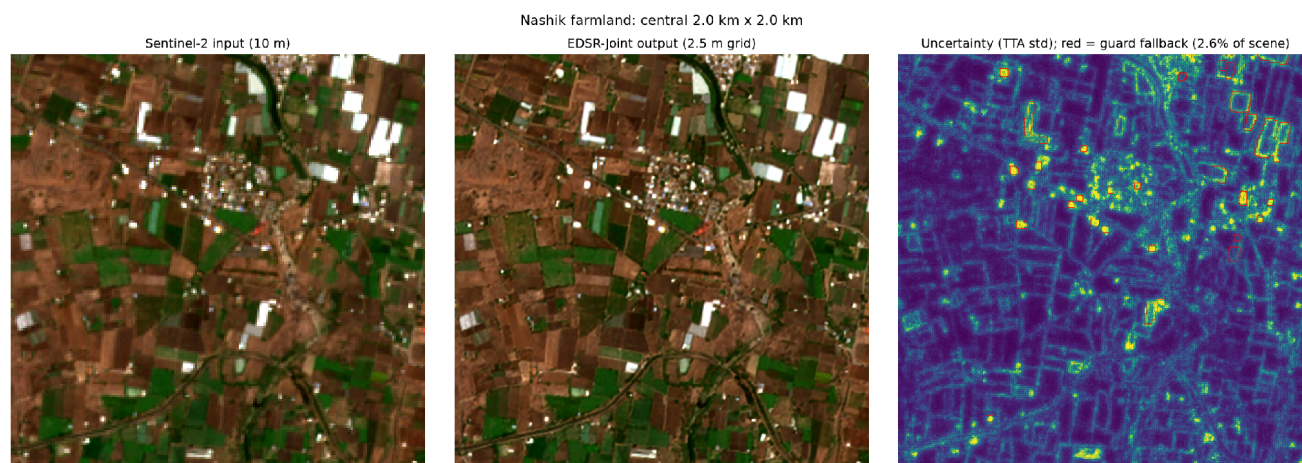


Figure 14. Nashik farmland (central 2 km × 2 km): parcels, polyhouses, roads and the river channel are sharpened; the guard (red, 2.6% of the scene) fires on bright polyhouse roofs.

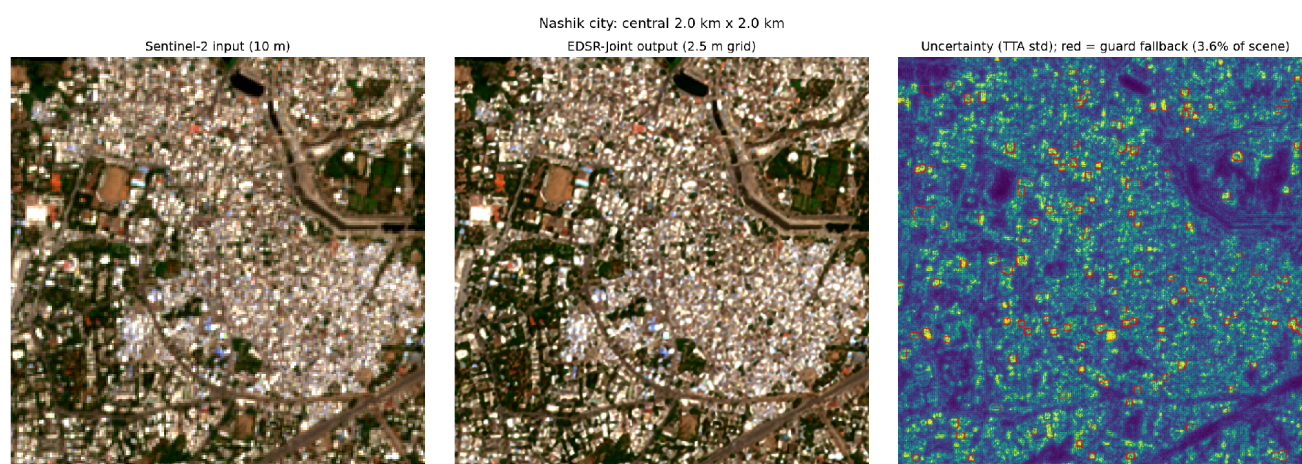


Figure 15. Nashik city (central 2 km × 2 km): roads, railway and open spaces sharpen, but dense old-city blocks stay smooth and are flagged as uncertain; guard fallback 3.6%.

6. RELEVANCE TO APPLICATIONS

- **Crop monitoring:** sharper field boundaries with NDVI error reduced by 19–22% versus bicubic on held-out Indian data, supporting parcel-level vegetation analysis.
- **Urban analysis:** roads, railways and open spaces are recovered; dense sub-10 m building fabric is not, and is flagged by the uncertainty map.
- **Water and disaster mapping:** river and reservoir edges are sharpened (Figure 14); the uncertainty and guard layers indicate where reconstructed detail should not be trusted for decisions.

Quantitative gains for these downstream tasks (e.g. water-mask IoU, field-boundary F1) were not measured and are listed as future work.

7. LIMITATIONS

1. **The <4 m claim is only partly supported.** The output grid is 2.5 m. On real 2.5 m references the model adds statistically significant real detail, but radiometric PSNR is on par with bicubic, and no 2.5 m reference was available over India. The product must not be treated as equivalent to native 2.5 m imagery.

2. **Synthetic training degradation.** Stage 1 uses a Gaussian PSF approximation of the Sentinel-2 MTF; stage 2 compensates with only 90 real pairs from the USA and Spain.
3. **Final-model Indian results come from a spatial hold-out** of the four training cities; unseen-city evidence (Nashik, Guwahati) was obtained with the stage-1 recipe.
4. **Domain coverage.** Four cities; humid terrain (Guwahati) gains less, and dense urban cores gain least on real data (Spain urban).
5. **Atmospheric quality.** Opaque clouds are masked, thin haze is not (Guwahati scene 5).
6. **Dynamic range.** Values above the training range are not super-resolved (guard fallback, 2–4% of pixels).
7. **Evaluation details.** Normalisation statistics include the test-city pixels (two numbers per band, no image content). Real-data results rest on 29 images; confidence intervals are reported for this reason.

8. FUTURE WORK

- Validate over India with licensed high-resolution references (e.g. Cartosat or aerial imagery) and extend fine-tuning with more real pairs (WorldStrat).
- Haze-aware screening (SCL thin-cirrus class, aerosol optical thickness) and more Indian cities from humid, coastal and hilly regions.
- Downstream evaluation: field-boundary F1, water-mask IoU and land-cover classification accuracy on SR vs 10 m inputs.
- Sensor-accurate MTF degradation, multi-date input fusion, and diffusion models once more real paired data and compute are available.

9. IMPLEMENTATION AND REPRODUCIBILITY

Table 10. Compute and implementation.

Item	Value
Hardware	NVIDIA GeForce RTX 3050 Laptop GPU (4.3 GB), Windows
Software	Python 3.11, PyTorch 2.6.0 (CUDA 12.4), rasterio 1.4.4 / GDAL, NumPy, pandas, QGIS for inspection
Model size	1.519 M parameters, 6.1 MB checkpoint
Stage-1 training	60 epochs, 0.75–1.14 h per run (41–79 s per epoch)
GAN ablation	15 epochs, 1.03 h
Stage-2 fine-tuning	30 epochs (wall-clock time not recorded)
Inference	10 km × 10 km scene → 2.5 m in 95.5 s with 8× TTA

The code base contains a data-integrity checker, training (stage 1, GAN, stage 2), synthetic and real-reference evaluation with bootstrap statistics, scene-level GeoTIFF inference, and figure generation from saved results, so every number in this report is traceable to a file. Main commands:

```
python scripts/train.py --config configs/edsr_baseline.yaml --set data.test_city=none "data.clip=[0,1.5]"
python scripts/finetune_opensr.py --ckpt runs/edsr_final_v2/best.pt --mix-synthetic 0.5 --out runs/edsr_ft_joint
python scripts/evaluate.py --ckpt runs/edsr_ft_joint/best.pt --split val --tta
python scripts/eval_opensr.py --ckpt runs/edsr_ft_joint/best.pt --datasets spot spain_urban --tta --norm-range 0 1.5
python scripts/infer.py --ckpt runs/edsr_ft_joint/best.pt --input scene_10m.tif --output scene_2p5m.tif --tta
```

Interactive tools. A Streamlit application runs the model live: it super-resolves an uploaded Sentinel-2 GeoTIFF (with uncertainty, guard and NDVI views and GeoTIFF download), and it re-runs the synthetic accuracy check on any held-out patch against its real 10 m truth. A web dashboard presents all benchmark results with confidence intervals and an interactive before/after view of the Nashik scene.

Repository: linked through the GitHub QR code in the submission deck. Data attribution: contains modified Copernicus Sentinel data (2025–2026); opensr-test / WorldStrat / NAIP references used under their respective licences.

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